

24 response summaries from 64 traceable wells

Measurement-aware drug-screen reconstruction under a fixed physical treatment-well budget

CURRENT DEV MSE

0.001058275

Bandwidth-0.7 additive

VS R13

-7.56%

same 24 targets

VS ADDITIVE

38 / 59

patient wins; 5/5 folds

Evidence status Retrospective repeated-development evidence on 119 Lib1 samples from 59 whole patients. Not independent prospective validation, not clinical guidance, and not an official competition score.

AI4S Open Innovation: AI for Life Science | Model & Algorithm

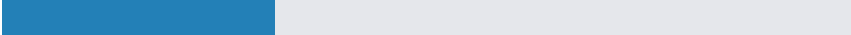
Team: von DosePilot | Author: Joseph Ayanda | Solo AI/ML/software submission

[Live demo](#) | [Source](#)

The contribution is a complete measurement-aware reconstruction system

DosePilot addresses a practical life-science bottleneck: when full dose-response profiling is expensive, which physical measurements should be purchased, and how much of the fixed response summary can those measurements reconstruct?

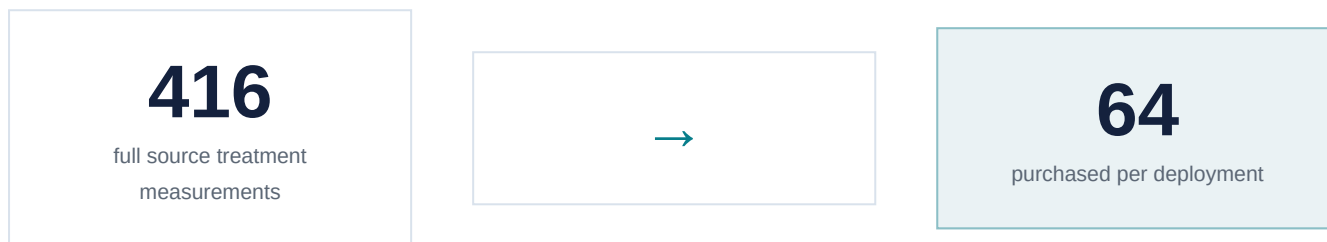
- A fixed 64-treatment schedule is committed before sample responses arrive.
- Each deployment consumes exactly 64 identified treatment measurements: 32 from each source plate.
- A training-contained own-drug ridge baseline is corrected by a cross-drug additive residual kernel.
- The frozen successor changes only one global Gaussian group lengthscale from 1.0 to 0.7; the physical plans stay identical.
- Whole-patient nested evaluation reports mean error, patient breadth, fold breadth, p90, and adverse target outcomes together.

R13		0.0011448587
S2		0.0010701439
Additive		0.0010605527
Bandwidth 0.7		0.0010582750

Current result MSE 0.001058275042: 7.56% below R13, 38/59 wins versus the immediate additive predecessor, all 5 outer folds improve, and p90 also improves. Ten of 24 target-average errors regress.

Sparse screening is an experimental-design problem, not just missing columns

Every DosePilot input corresponds to a real drug, exact native dose, source plate, and treatment well. A model that silently substitutes a concentration, ensembles both orientations, or fills a missing value changes the physical experiment.



15.38% of the source treatment-measurement count. This is not a 84.62% laboratory cost/time claim.

- Full selected source curves contain 416 treatment measurements per sample (208/plate).
- The frozen policy selects 2 doses for all 24 drugs plus 16 third-dose upgrades: $24 \times 2 + 16 = 64$.
- Controls are separate declared resources outside the 64 treatment-well budget.
- The project therefore claims measurement-count compression, not a direct cost, labor, chip-resource, or elapsed-time percentage.

Life-science value A reliable sparse reconstruction layer can help prioritize limited assay capacity and create an auditable bridge from model selection to a prospective organ-on-chip protocol. Actual laboratory utility still requires a new study.

The 24 outputs are fixed normalized log-dose AUC summaries

For drug target j and source plate r , the reference is the exact trapezoidal integral of supplied normalized viability over a fixed log-dose interval, with linear interpolation in log-dose at a boundary when needed.

$$\begin{aligned} \text{AUC}_{j,r} &= \int v_{j,r}(x) \, dx / [\log(U_j) - \log(L_j)] \\ \text{target}_j &= \frac{1}{2}(\text{AUC}_{j,p1} + \text{AUC}_{j,p2}) \end{aligned}$$

- Supplied viability values are not clipped before integration.
- The output is dimensionless normalized log-dose AUC, not IC50, drug rank, or a clinical response label.
- Purchased measurements may also be nodes contributing to the measured reference AUC; this is reconstruction of a measured full-profile summary, not recovery of noiseless truth.
- A/B are input-layout alternatives. The target definition itself always averages the full p1 and p2 reference AUCs.

Boundary interpolation Gedatolisib, Palbociclib and TAS-102 use an interpolated lower integration bound; SN-38 uses an interpolated upper bound. All dose-only quadrature weights are public and sum to 1.

Frozen predictor doses, targets 1–12

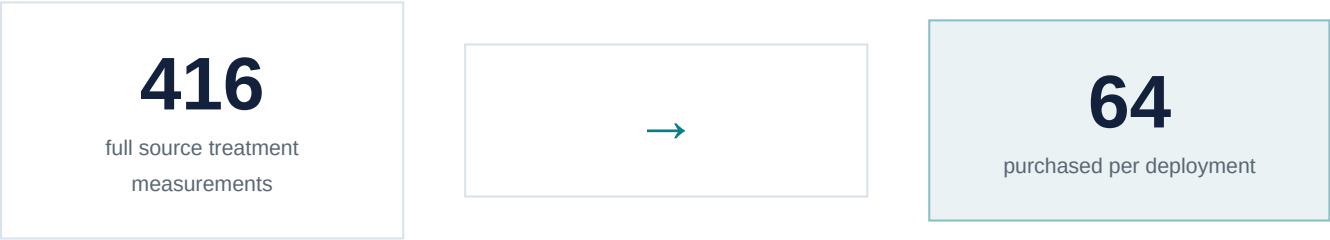
Target	Frozen predictor doses (nM)	Dose count
5-FU	1000, 3000	2
AZD7762	100, 300	2
Afatinib	10, 100, 300	3
Alisertib	100, 1000, 10000	3
Atorvastatin	300, 1000, 3000	3
Bemcentinib	300, 3000	2
Encorafenib	30, 300	2
Gedatolisib	30, 100, 300	3
Gemcitabine	10, 100, 300	3
Idasanutlin	300, 1000, 3000	3
LCL161	250, 750, 25000	3
LGK974	10, 1000, 3000	3

The final construction pins the original acquisition plan. Dose identities are fixed independently of the new bandwidth parameter.

Orientation contract A and B contain the same 64 drug+dose identities. Their source-plate assignments are complementary. Each orientation alone has 32 p1 + 32 p2 measurements.

Frozen predictor doses, targets 13–24

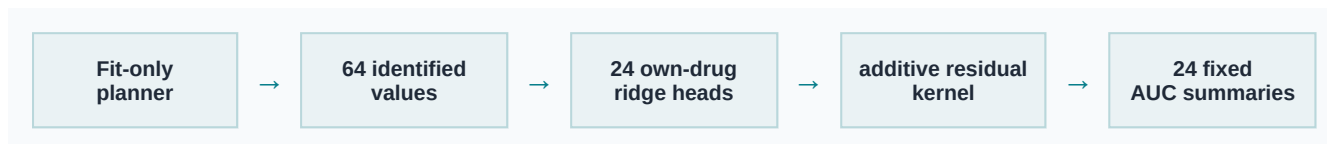
Target	Frozen predictor doses (nM)	Dose count
Lapatinib	30, 100, 300	3
Luminespib	10, 30, 300	3
Methotrexate	150, 500, 1500	3
Napabucasin	600, 2000	2
Palbociclib	300, 1000, 3000	3
Panobinostat	10, 30, 100	3
Pevonedistat	300, 1000	2
Regorafenib	300, 3000	2
SN-38	3, 100, 300	3
TAS-102	100, 1000, 10000	3
Trametinib	2.5, 7.5, 75	3
Volasertib	10, 100	2



15.38% of the source treatment-measurement count. This is not a 84.62% laboratory cost/time claim.

Exactly what the predictor receives One orientation supplies the 64 raw purchased normalized-viability values. It does not receive the unpurchased full curves, the 24 target AUCs, the other orientation, a 128-well A+B ensemble, or Protected22/Lib2 responses.

Training-contained acquisition, own-drug reconstruction, then additive residual correction



Stage 1 — acquisition

For each drug, fitting-only moments compare native two-dose and three-dose subsets with planning penalty $\alpha=0.1$. Sixteen third-dose upgrades are allocated by estimated reduction in regularized reconstruction variance.

$$\text{subset score} = \text{Var}(y) - c^T (G + \alpha I)^{-1} c$$

Stage 2 — own-drug ridge

Each target sees only its own two or three purchased coordinates. Means and scales are refit in every fitting split, with the historical 0.05 scale floor and base ridge $\lambda=0.01$.

Stage 3 — bandwidth-0.7 additive residual kernel

$$k_j(z, z') = d_j \cdot \exp[-||z_j - z'_j||^2 / (2 d_j \cdot 0.7^2)]$$

The complete residual kernel is the sum of 24 drug-group Gaussian components plus the unchanged linear component. The 0.7 successor changes only this global group lengthscale.

Nested whole-patient containment and explicit A/B semantics

Outer evaluation

5 whole-patient folds

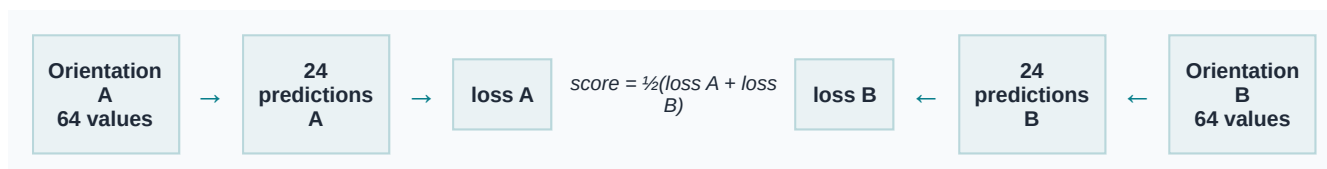
Every sample from one patient stays on one side of the split. The complete procedure — acquisition, means, scales, model fit, and hyperparameter choice — is rebuilt without the held patient.

Inner selection

3 whole-patient folds

One common residual spectral option is selected across all 24 targets. Candidate options are identity plus fraction $\{0.1, 0.3, 0.6\} \times \text{ridge } \{0.1, 1, 10\}$.

A and B are alternatives, not an ensemble



- A and B are separately budget-valid 64-well deployments.
- Losses are averaged to estimate expected orientation performance. Prediction vectors are never averaged.
- The 238 runtime requests are 119 samples $\times$ 2 alternative orientations; they are not 238 independent biological samples.

Promotion discipline A successor must lower mean error, improve enough patient means and outer folds, keep p90 nonworse, and continue to pass the historical R13/R18 screen. A lower decimal alone is insufficient.

The current bandwidth successor improves the same-task frontier without changing one well



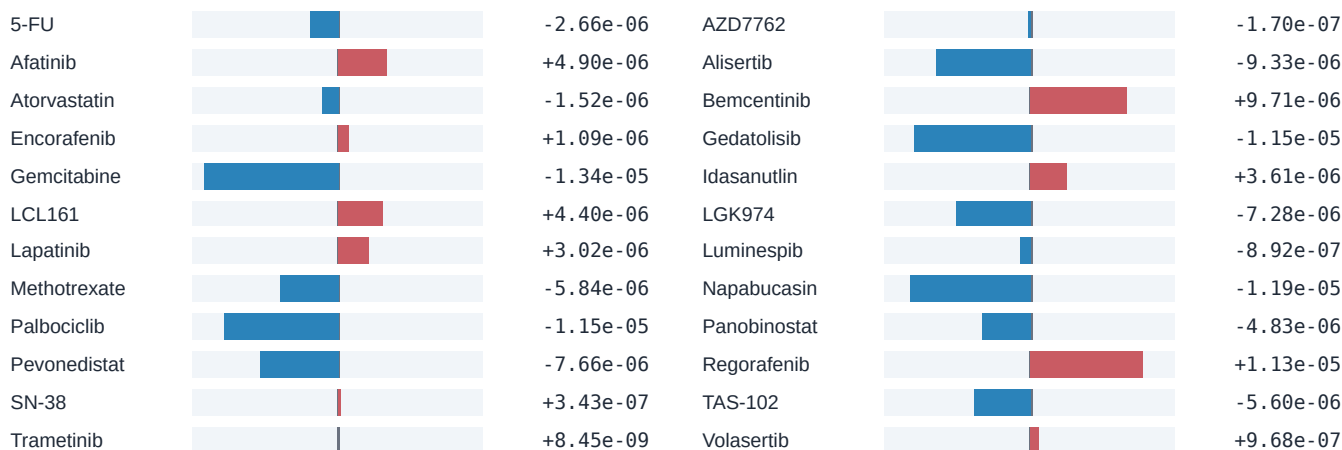
MSE	P90 RMSE	VS ADDITIVE
0.001058275	0.0378943	38 / 59
patient-balanced full24	vs 0.0380731 additive	patient wins; 5 / 5 folds

Comparison	Relative MSE gain	Patient wins	Favorable folds
vs original R13	7.56%	49 / 59	5 / 5
vs archived R18	7.28%	47 / 59	5 / 5
vs additive 1.0	0.215%	38 / 59	5 / 5

The two orientation-wide MSEs are 0.0011047522 and 0.0010117979. The descriptive paired-patient interval for current-minus-additive mean loss is $[-4.159e-6, -4.157e-7]$. It is not selection-corrected because the development population has been reused across research rounds.

Interpretation The large gain is from the complete progression versus R13; the final 0.7 bandwidth step is deliberately modest. Its value is that it improves mean, patient breadth, all five folds, and p90 with identical physical plans.

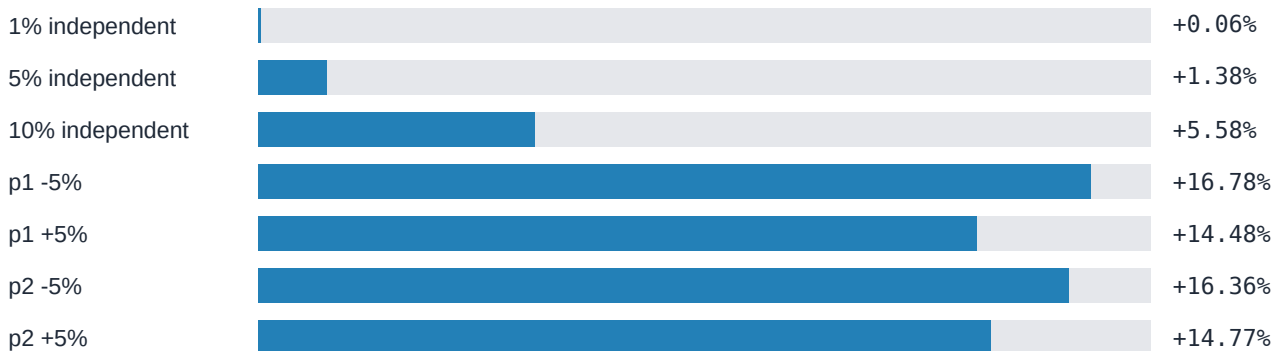
Fourteen targets improve or tie; ten target means regress



Negative bars improve. The largest improvements include Gemcitabine, Napabucasin, Gedatolisib and Palbociclib. The largest regressions include Regorafenib, Bemcentinib and Afatinib.

Why this stays in the headline package A project can improve aggregate performance while becoming worse for scientifically important slices. DosePilot therefore publishes all 24 target deltas and does not describe the current model as uniformly superior.

Random input noise is modest; coherent plate drift is the sharper failure mode



The frozen post-hoc stress test perturbs already-fitted outer-fold models with no refitting or model selection. Independent Gaussian input noise is expressed in model z-space and averaged over five fixed seeds.

A coherent $\pm 5\%$ multiplicative offset on one source plate is much more damaging than modest independent noise, increasing MSE by roughly 14.5%–16.8%. This is a sensitivity warning, not an estimate of real assay drift.

Missing measurement policy A required missing purchased reading withholds all 24 primary bandwidth predictions. Existing runtime tests checked every one of the 64 single-missing positions. No value is silently imputed and no free replacement well is assumed.

Public source-to-result replay, identity-bound inference, and durable evidence

- The exact public Mendeley Data v3 workbook and the reconstructed TRAIN CSV are SHA-256 pinned.
- Fresh-environment public replay reproduces bandwidth-0.7 MSE 0.0010582750420801538, additive 0.001060552730112811, and R13 0.001144858681382854 within 1e-12.
- No historical prediction file or old private metadata kit is an input to the bandwidth public replay.
- Runtime verification covers 238 sample/orientation requests and matches direct matrix evaluation within 2.22e-16.
- The constructed model binds native IDs, drug IDs, exact concentrations, plate orientation, model/plan hashes, and requires all 64 finite values.
- The durable lifecycle uses commit → optional baseline-only recovery → predict, create-exclusive evidence files, and same-frame serialization on tested local POSIX filesystems.

```
python study/hybrid_residual/reproduce_bandwidth.py --curves reconstructed_train/train_curves.csv --output bandwidth_replay --fit-final
```

Scope of software evidence Numerical parity, hash binding, identity checks, and durable file behavior demonstrate software correctness under tested conditions. They do not certify physical execution, biological adequacy, disk-level power-loss behavior, or prospective predictive validity.

External tests are reported as support, stress tests, or non-estimable outcomes

Assessment	Frame	Point comparison	Gate outcome
FORECAST-1	13/19 complete confirmation patients	0.0021715 vs 0.0038552 interpolation	10/13 wins; original support gate failed
Matched tumor-CAF	15 matched coculture cases	0.0029697 vs 0.0052366 interpolation	10/15 wins; p90 0.06666 vs 0.10986; gate passed
eLife stress test	12 organoids, 5-drug adapted task	0.0041734 vs 0.0060423 interpolation	7/12 wins; gate failed
Protected22 / Lib2	22 supported heads	Full primary not estimable	5 required responses nonnumeric; prior exposure

These sources do not transform the current bandwidth-0.7 Lib1 result into independent validation. They answer narrower questions about adapted sparse-reconstruction designs, context shifts, stronger controls, and operational failure modes.

Selection-aware wording The defensible claim is: DosePilot has a strong, reproducible retrospective development result plus mixed external/contextual evidence. This report does not claim prospective organ-on-chip validation or clinical utility.

The treatment schedule is frozen; device-specific biology is intentionally still TBD

The final bandwidth construction pins the 64-treatment acquisition plan. Public orientation-A and orientation-B schedules fix drug identity, exact dose, and source-plate assignment before any future response collection.

- Each orientation contains 64 distinct treatment identities, exactly 32 from p1 and 32 from p2.
- A and B contain the same drug+dose identities with complementary source-plate assignments.
- All 24 targets are represented: 8 with two doses and 16 with three.
- Binding templates leave device, compartment, circuit, reservoir, channel, dosing route, exposure duration, readout timepoint and readout type as TBD_BEFORE_PROSPECTIVE_COLLECTION.
- A response-free verifier checks schedule hashes, counts, A/B complementarity, target coverage, template integrity, public-site parity and separate control policy.

Frozen schedule property	A	B
Distinct treatment rows	64	64
Source-plate p1 rows	32	32
Source-plate p2 rows	32	32
Two-dose targets	8	8
Three-dose targets	16	16

Compiler scope A synthetic one-independent-circuit-per-treatment witness compiles to 64 treatment actions plus two separate controls. That proves encoded logical consistency with an invented witness only; it does not establish device availability, tissue compatibility, flow adequacy, or biological validity.

A prospective study contract is specified before the first new response

A genuinely confirmatory organ-on-chip study must use new biological units and freeze the model, 64-treatment schedule, device manifest, target definition, QC rules, statistical analysis, and reference assay before response collection. Exposed Lib2/Protected22 records are ineligible as untouched confirmation.

Before collection	Required commitment
Biological unit	Independent patient-level unit; repeated technical samples remain grouped.
Device	Platform, material, compartments, circuits/reservoirs, flow conditions, dosing route.
Assay	Exposure duration, readout technology, timepoint, normalization, exact control layout.
QC	Plate-level drift estimator plus acceptance/rejection or correction rule fixed before target reveal.
Reference	Full-profile target assay, dose grid, integration, failed-measurement policy, fixed scoring mask.
Prediction	Commit model hash + identified 64-value file, predict and seal output, then reveal reference targets.
Statistics	Paired patient-level primary analysis, sample-size/power plan, confidence procedure, adverse target reporting.

Recommended primary confirmatory question: Does frozen bandwidth-0.7 have lower patient-balanced full-24 MSE than frozen R13 on genuinely new biological units under the same prespecified scoring mask?

Unresolved by design Exact chip platform, tissue preparation/passage, exposure time, readout, control layout, numeric plate-QC threshold, reference assay, prospective cohort size/power, confidence procedure, and rerun policy must be closed before a real experiment.

The incumbent survived bounded challengers; failures remain public

Method / branch	MSE	Decision
R13 own-drug coverage	0.0011448587	Historical retained baseline
S2 spectral residual	0.0010701439	Promoted development successor
Additive drug-group kernel	0.0010605527	Previous incumbent
Bandwidth 0.7	0.0010582750	Current incumbent; passes incumbent + historical screens
Bandwidth 1.4	0.0010637094	Rejected: 18/59 wins, 0/5 folds
Residual-alignment reweighting	0.0010608378	Rejected: worse mean/p90, 2/5 folds
Multioutput acquisition sweep	0.0010665434	Rejected: 21/59 wins, 2/5 folds, worse p90
Iterated additive, 2 cycles	0.0010693535	Rejected
A/B consistency regularization	~0.001080+	Rejected
Matérn nonlinear challenger	0.0011461852	Rejected vs matching control

Many additional exploratory branches remain in project history. The public package keeps enough negative evidence to distinguish a controlled research program from a single cherry-picked score.

Promotion rule Against the immediate incumbent a candidate must lower mean MSE, improve at least 30/59 patient means and 3/5 outer folds, keep p90 nonworse, and continue to pass the historical R13/R18 screen.

What the current evidence does and does not establish

The strongest limitation is project-level selection reuse. Individual experiments use patient-grouped containment, but many hypotheses have been inspected on the same 59-patient development population. The current score is therefore a selected development estimate until genuinely new prospective units are evaluated under a frozen protocol.

Evidence level	What is demonstrated	What is not demonstrated
Development prediction	Nested whole-patient reconstruction accuracy on Lib1; same-plan bandwidth improvement; adverse target outcomes.	Untouched confirmation, calibrated uncertainty, clinical utility.
Software execution	Public-input replay, identity/hash binding, 64-well accounting, missing-input withholding, durable lifecycle.	Physical assay execution, device availability, wet-lab QC performance.
External/contextual	One matched-CAF adapted gate pass; other stress/support outcomes including failures.	Independent validation of the exact current fitted bandwidth model.
Prospective OoC readiness	Frozen 64-row schedules, binding templates, compiler checks, prospective validation contract.	Prospective organ-on-chip results or measured laboratory savings.

- The measured endpoint can reuse purchased nodes; reconstruction accuracy is not prediction of an entirely independent biological truth.
- The 64/416 ratio is treatment-measurement count compression, not a direct cost/time/chip-resource reduction.
- Controls are separate resources and must be accounted for in a real assay.
- Plate-drift simulation suggests coherent calibration error is more dangerous than modest independent noise, but no numeric QC threshold is validated.
- Ten target-average errors regress versus the immediate additive predecessor.
- Current evidence supports research software and a prospective study plan, not clinical decision-making.

A reviewer can replay the core result from the public source route

Minimal scientific replay from repository root:

```
python -m venv .study-venv
python -m pip install -r study/requirements.txt
python study/acquire_public_source.py --output Data_S4.xlsx
python study/prepare_compact_source.py --source-xlsx Data_S4.xlsx --output reconstructed_train --
execute-lib1-only
python study/hybrid_residual/reproduce_bandwidth.py --curves reconstructed_train/train_curves.csv --
output bandwidth_replay --fit-final
python study/audits/verify_finalist_audit.py --root .
python study/audits/verify_frozen_ooc_schedule.py --root evidence --repo .
```

Data and licensing. Original DosePilot code and documentation are MIT licensed. External datasets and papers retain their own terms. The project does not redistribute the biological source workbook, patient arrays, or fitted biological weights. The primary Lib1 source is Kryeziu et al. (2026) and Mendeley Data v3, listed CC BY 4.0.

AI assistance. AI coding/research assistants supported development. Public numerical claims are tied to executable source, hashes, aggregate receipts, and response-free consistency tests rather than accepted from generated prose alone.

Primary references

1. Kryeziu et al. (2026). *Cell Reports Medicine*. DOI 10.1016/j.xcrm.2026.102840.
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5. Tan et al. (2023). *Cell Reports Medicine*. DOI 10.1016/j.xcrm.2023.101335.
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[Repository](#) | [Live demo](#) | [Original demo video](#)

Bottom line DosePilot demonstrates a reproducible, measurement-aware reconstruction system under an explicit 64-treatment budget, with a small but broad matched-model improvement at bandwidth 0.7 and a concrete, audited handoff to a future prospective organ-on-chip study. Development accuracy, software execution, and biological validation remain explicitly separated.